THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A right circular cone has base radius 14 cm and height 48 cm. The curved surface area of the cone is: 1

(a) 616 cm² **(b)** 2112 cm² **(c)** 2816 cm² **(d)** 2200 cm²

2. A solid sphere has radius 14 cm. The surface area of the sphere is: 1

(a) 1848 cm² **(b)** 2464 cm² **(c)** 616 cm² **(d)** 1232 cm²

3. A solid sphere has radius 42 cm. The volume of the sphere is: 1

(a) 155232 cm³ **(b)** 232848 cm³ **(c)** 310464 cm³ **(d)** 22176 cm³

4. A solid hemisphere has radius 14 cm. The total surface area of the hemisphere is: 1

(a) 1848 cm² **(b)** 616 cm² **(c)** 2464 cm² **(d)** 1232 cm²

5. A right circular cylinder has base radius 21 cm and height 8 cm. The curved surface area of the cylinder is: 1

(a) 3828 cm² **(b)** 1056 cm² **(c)** 11088 cm² **(d)** 2112 cm²

6. A right circular cylinder has base radius 21 cm and height 2 cm. The volume of the cylinder is: 1

(a) 3036 cm³ **(b)** 2772 cm³ **(c)** 264 cm³ **(d)** 924 cm³

7. A right circular cone has base radius 21 cm and height 28 cm. The slant height of the cone is: 1

(a) 49 cm **(b)** 24.5 cm **(c)** 27 cm **(d)** 35 cm

8. A right circular cone has base radius 21 cm and height 28 cm. The curved surface area of the cone is: 1

(a) 1848 cm² **(b)** 3696 cm² **(c)** 2310 cm² **(d)** 1386 cm²

9. A solid sphere has radius 21 cm. The surface area of the sphere is: 1

(a) 2772 cm² **(b)** 4158 cm² **(c)** 1386 cm² **(d)** 5544 cm²

10. A solid sphere has radius 10.5 cm. The volume of the sphere is: 1

(a) 1386 cm³ **(b)** 4851 cm³ **(c)** 3638.25 cm³ **(d)** 2425.5 cm³

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid hemisphere has radius 21 cm. Find the volume of the hemisphere. 2

12. A cuboid measures 12 cm by 7 cm by 4 cm. Find the total surface area of the cuboid. 2

13. A right circular cylinder has base radius 21 cm and height 4 cm. Find the total surface area of the cylinder. 2

14. A solid hemisphere has radius 31.5 cm. Find the volume of the hemisphere. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A right circular cone has base radius 21 cm and height 28 cm. Find the volume of the cone. 3

16. A right circular cone has base radius 21 cm and height 28 cm. Find the total surface area of the cone. 3

17. The total surface area of a cube is 150 cm^2. Find the volume of the cube. 3

OR

A right circular cone has base radius 21 cm and height 72 cm. Find the volume of the cone. 3

18. A right circular cone has base radius 21 cm and height 72 cm. Find the total surface area of the cone. 3

OR

The total surface area of a cube is 216 cm^2. Find the volume of the cube. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A solid metal cuboid 24 cm by 18 cm by 9 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 12 cm is melted and recast into small solid spheres of radius 4 cm. Find the number of small spheres obtained. 5

20. A solid metal cuboid 24 cm by 18 cm by 18 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 10 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres obtained. 5

END OF QUESTION PAPER — Total: 40 Marks

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